

Лабораторная работа №4

Первоначальное конфигурирование сети

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Цель работы

Провести подготовительную работу по первоначальной настройке коммутаторов сети в Cisco Packet Tracer.

Построение сети L1

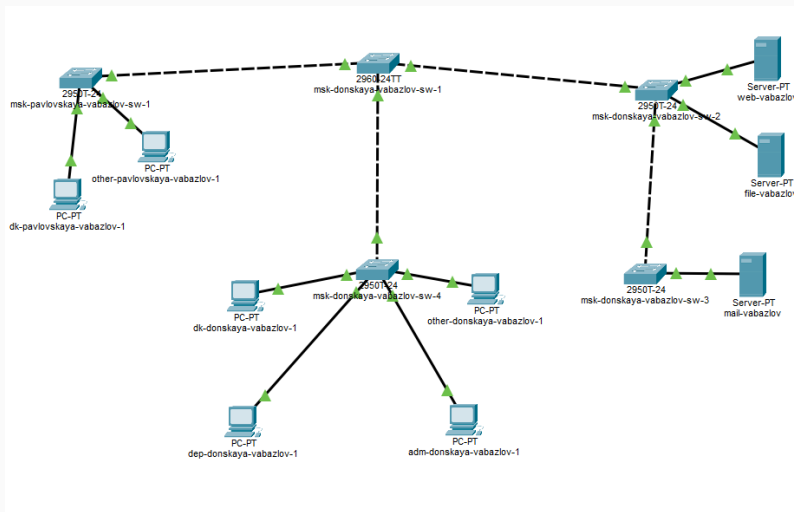


Рис. 1: Топология сети L1

Первоначальная настройка

Для каждого коммутатора выполнены:

- назначение hostname
- настройка интерфейса VLAN 2
- назначение IP-адреса
- указание шлюза по умолчанию
- настройка паролей
- создание пользователя admin
- включение SSH
- сохранение конфигурации

Настройка msk-donskaya-vabazlov-sw-1

```
msk-donskaya-vabazlov-sw-1

Physical  Config  CLI  Attributes

IOS Command Line Interface

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#host msk-donskaya-vabazlov-sw-1
msk-donskaya-vabazlov-sw-1(config)#int vlan2
msk-donskaya-vabazlov-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan2, changed state to up

msk-donskaya-vabazlov-sw-1(config-if)#no sh
msk-donskaya-vabazlov-sw-1(config-if)#ip addr 10.128.1.2 255.255.255.0
msk-donskaya-vabazlov-sw-1(config-if)#exit
msk-donskaya-vabazlov-sw-1(config)#ip default-gate 10.128.1.1
msk-donskaya-vabazlov-sw-1(config)#line vty 0 4
msk-donskaya-vabazlov-sw-1(config-line)#pass cisco
msk-donskaya-vabazlov-sw-1(config-line)#login
msk-donskaya-vabazlov-sw-1(config-line)#exit
msk-donskaya-vabazlov-sw-1(config)#line cons 0
msk-donskaya-vabazlov-sw-1(config-line)#pass cisco
msk-donskaya-vabazlov-sw-1(config-line)#login
msk-donskaya-vabazlov-sw-1(config-line)#exit
msk-donskaya-vabazlov-sw-1(config)#enab sec cisco
msk-donskaya-vabazlov-sw-1(config)#serv pass
msk-donskaya-vabazlov-sw-1(config)#user admin priv 1 sec cisco
msk-donskaya-vabazlov-sw-1(config)#ip domain-name donskeya.rudn.edu
msk-donskaya-vabazlov-sw-1(config)#crypto key gen rsa
The name for the keys will be: msk-donskaya-vabazlov-sw-1.donskeya.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

msk-donskaya-vabazlov-sw-1(config)#line vty 0 4
*Mar 1 0:9:30.507: RSA key size needs to be at least 768 bits for ssh version 2
```


Настройка msk-donskaya-vabazlov-sw-2

IOS Command Line Interface

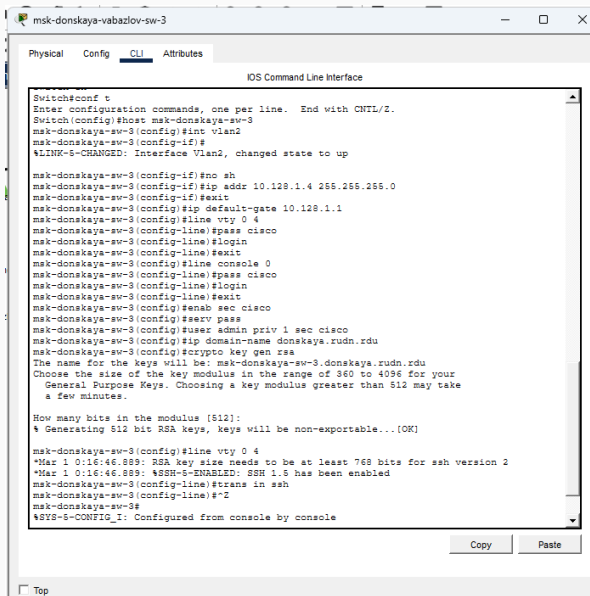
```
msk-donskaya-vabazlov-sw-2(config-if)#no shut
msk-donskaya-vabazlov-sw-2(config-if)#ip addr 10.128.1.3 255.255.255.0
msk-donskaya-vabazlov-sw-2(config-if)#exit
msk-donskaya-vabazlov-sw-2(config)#ip default-gate 10.128.1.1
msk-donskaya-vabazlov-sw-2(config)#line vty 0 4
msk-donskaya-vabazlov-sw-2(config-line)#pass cisco
msk-donskaya-vabazlov-sw-2(config-line)#login
msk-donskaya-vabazlov-sw-2(config-line)#exit
msk-donskaya-vabazlov-sw-2(config)#line console 0
msk-donskaya-vabazlov-sw-2(config-line)#pass cisco
msk-donskaya-vabazlov-sw-2(config-line)#login
msk-donskaya-vabazlov-sw-2(config-line)#exit
msk-donskaya-vabazlov-sw-2(config)#en sec cisco
% Ambiguous command: "en sec cisco"
msk-donskaya-vabazlov-sw-2(config)#enab sec cisco
msk-donskaya-vabazlov-sw-2(config)#serv pass
msk-donskaya-vabazlov-sw-2(config)#user admin priv 1 sec cisco
msk-donskaya-vabazlov-sw-2(config)#ip domain-name donsкаya.rudn.edu
msk-donskaya-vabazlov-sw-2(config)#crypto key gen rsa
The name for the keys will be: msk-donskaya-vabazlov-sw-2.donsкаya.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

msk-donskaya-vabazlov-sw-2(config)#line vty 0 4
*Mar 1 0:13:34.410: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 0:13:34.410: %SSH-5-ENABLED: SSH 1.5 has been enabled
msk-donskaya-vabazlov-sw-2(config-line)#trans in ssh
msk-donskaya-vabazlov-sw-2(config-line)#^Z
msk-donskaya-vabazlov-sw-2#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-vabazlov-sw-2#wr mem
Building configuration...
[OK]
msk-donskaya-vabazlov-sw-2#
```

Настройка msk-donskaya-vabazlov-sw-3



The screenshot shows a network configuration window titled "msk-donskaya-vabazlov-sw-3". It has tabs for "Physical", "Config", "CLI", and "Attributes". The "CLI" tab is active, displaying the "IOS Command Line Interface". The interface shows a series of configuration commands and their outputs for a switch named "msk-donskaya-sw-3".

```
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#host msk-donskaya-sw-3
msk-donskaya-sw-3(config)#int vlan2
msk-donskaya-sw-3(config-if)#
%LINK-5-CHANGED: Interface Vlan2, changed state to up

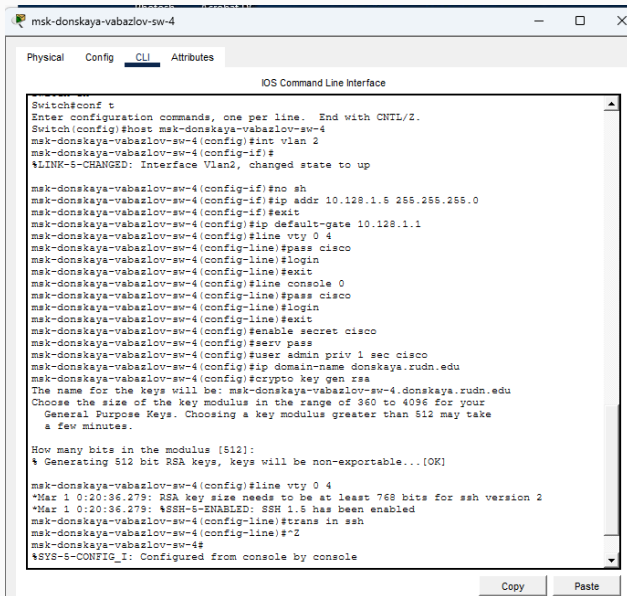
msk-donskaya-sw-3(config-if)#no sh
msk-donskaya-sw-3(config-if)#ip addr 10.128.1.4 255.255.255.0
msk-donskaya-sw-3(config-if)#exit
msk-donskaya-sw-3(config)#ip default-gate 10.128.1.1
msk-donskaya-sw-3(config)#line vty 0 4
msk-donskaya-sw-3(config-line)#pass cisco
msk-donskaya-sw-3(config-line)#login
msk-donskaya-sw-3(config-line)#exit
msk-donskaya-sw-3(config)#line console 0
msk-donskaya-sw-3(config-line)#pass cisco
msk-donskaya-sw-3(config-line)#login
msk-donskaya-sw-3(config-line)#exit
msk-donskaya-sw-3(config)#enab sec cisco
msk-donskaya-sw-3(config)#serv pass
msk-donskaya-sw-3(config)#user admin priv 1 sec cisco
msk-donskaya-sw-3(config)#ip domain-name donskeya.rudn.rdu
msk-donskaya-sw-3(config)#crypto key gen rsa
The name for the keys will be: msk-donskaya-sw-3.donskeya.rudn.rdu
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

msk-donskaya-sw-3(config)#line vty 0 4
*Mar 1 0:16:46.889: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 0:16:46.889: %SSH-5-ENABLED: SSH 1.5 has been enabled
msk-donskaya-sw-3(config-line)#trans in ssh
msk-donskaya-sw-3(config-line)#-Z
msk-donskaya-sw-3#
%SYS-5-CONFIG_I: Configured from console by console
```

At the bottom right of the CLI window, there are "Copy" and "Paste" buttons. Below the window, there is a "Top" button with a small square icon to its left.

Настройка msk-donskaya-vabazlov-sw-4



The screenshot shows a network configuration window titled "msk-donskaya-vabazlov-sw-4". It has tabs for "Physical", "Config", "CLI", and "Attributes". The "CLI" tab is active, displaying the "IOS Command Line Interface". The interface shows a series of configuration commands and their outputs:

```
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#host msk-donskaya-vabazlov-sw-4
msk-donskaya-vabazlov-sw-4(config)#int vlan 2
msk-donskaya-vabazlov-sw-4(config-if)#
%LINK-5-CHANGED: Interface Vlan2, changed state to up

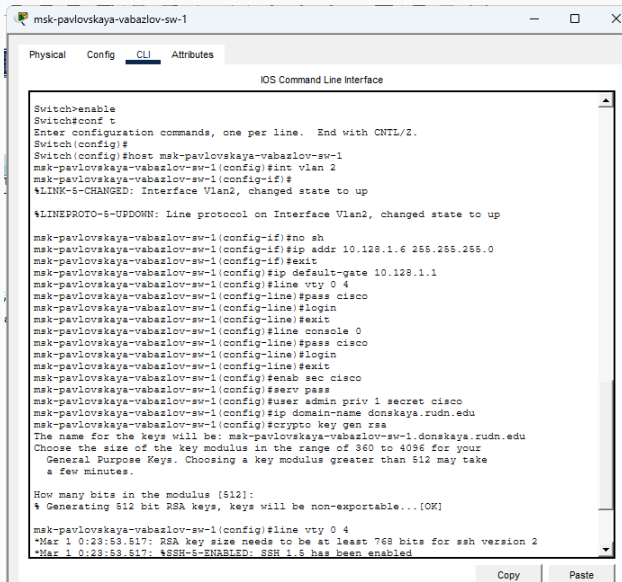
msk-donskaya-vabazlov-sw-4(config-if)#no sh
msk-donskaya-vabazlov-sw-4(config-if)#ip addr 10.128.1.5 255.255.255.0
msk-donskaya-vabazlov-sw-4(config-if)#exit
msk-donskaya-vabazlov-sw-4(config)#ip default-gate 10.128.1.1
msk-donskaya-vabazlov-sw-4(config)#line vty 0 4
msk-donskaya-vabazlov-sw-4(config-line)#pass cisco
msk-donskaya-vabazlov-sw-4(config-line)#login
msk-donskaya-vabazlov-sw-4(config-line)#exit
msk-donskaya-vabazlov-sw-4(config)#line console 0
msk-donskaya-vabazlov-sw-4(config-line)#pass cisco
msk-donskaya-vabazlov-sw-4(config-line)#login
msk-donskaya-vabazlov-sw-4(config-line)#exit
msk-donskaya-vabazlov-sw-4(config)#enable secret cisco
msk-donskaya-vabazlov-sw-4(config)#serv pass
msk-donskaya-vabazlov-sw-4(config)#user admin priv 1 sec cisco
msk-donskaya-vabazlov-sw-4(config)#ip domain-name donskeya.rudn.edu
msk-donskaya-vabazlov-sw-4(config)#crypto key gen rsa
The name for the keys will be: msk-donskaya-vabazlov-sw-4.donskeya.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

msk-donskaya-vabazlov-sw-4(config)#line vty 0 4
*Mar 1 0:20:36.279: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 0:20:36.279: %SSH-5-ENABLED: SSH 1.5 has been enabled
msk-donskaya-vabazlov-sw-4(config-line)#trans in ssh
msk-donskaya-vabazlov-sw-4(config-line)#*2
msk-donskaya-vabazlov-sw-4#
%SYS-5-CONFIG_I: Configured from console by console
```

At the bottom of the window, there are "Copy" and "Paste" buttons.

Настройка msk-pavlovskaya-vabazlov-sw-1



The screenshot shows a network configuration window titled "msk-pavlovskaya-vabazlov-sw-1". It has tabs for "Physical", "Config", "CLI", and "Attributes", with "CLI" selected. The main area is titled "IOS Command Line Interface" and displays a series of commands and their outputs. The commands configure the switch, set the host name, create VLAN 2, configure the interface, set IP address, default gateway, and enable SSH.

```
Switch>enable
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#
Switch(config)#host msk-pavlovskaya-vabazlov-sw-1
msk-pavlovskaya-vabazlov-sw-1(config)#int vlan 2
msk-pavlovskaya-vabazlov-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan2, changed state to up

msk-pavlovskaya-vabazlov-sw-1(config-if)#no sh
msk-pavlovskaya-vabazlov-sw-1(config-if)#ip addr 10.128.1.6 255.255.255.0
msk-pavlovskaya-vabazlov-sw-1(config-if)#exit
msk-pavlovskaya-vabazlov-sw-1(config)#ip default-gate 10.128.1.1
msk-pavlovskaya-vabazlov-sw-1(config)#line vty 0 4
msk-pavlovskaya-vabazlov-sw-1(config-line)#pass cisco
msk-pavlovskaya-vabazlov-sw-1(config-line)#login
msk-pavlovskaya-vabazlov-sw-1(config-line)#exit
msk-pavlovskaya-vabazlov-sw-1(config)#line console 0
msk-pavlovskaya-vabazlov-sw-1(config-line)#pass cisco
msk-pavlovskaya-vabazlov-sw-1(config-line)#login
msk-pavlovskaya-vabazlov-sw-1(config-line)#exit
msk-pavlovskaya-vabazlov-sw-1(config)#enab sec cisco
msk-pavlovskaya-vabazlov-sw-1(config)#serv pass
msk-pavlovskaya-vabazlov-sw-1(config)#user admin priv 1 secret cisco
msk-pavlovskaya-vabazlov-sw-1(config)#ip domain-name donskaya.rudn.edu
msk-pavlovskaya-vabazlov-sw-1(config)#crypto key gen rsa
The name for the keys will be: msk-pavlovskaya-vabazlov-sw-1.donskaya.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

msk-pavlovskaya-vabazlov-sw-1(config)#line vty 0 4
*Mar 1 0:23:53.517: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 0:23:53.517: %SSH-5-ENABLED: SSH 1.5 has been enabled
```

At the bottom right, there are "Copy" and "Paste" buttons.

Итоги работы

В ходе лабораторной работы:

- построена сеть L1;
- выполнена базовая настройка коммутаторов;
- назначены IP-адреса управления;
- реализована защита доступа;
- настроен удалённый доступ по SSH;
- конфигурация сохранена в NVRAM.

Сеть подготовлена к дальнейшему конфигурированию и тестированию.